

TypeOps Score

Hackathon Idea Brief

An automated brand typography health score for enterprise teams

Why a Score Changes Everything

The most powerful business innovations are often not products — they are measurements. When you give people a number, you give them a problem they can act on. Two examples prove this beyond doubt.

The NPS story: one number that transformed customer experience

Before NPS (pre-2003)

Companies ran lengthy customer satisfaction surveys — 20 to 40 questions, weeks to process, results nobody could agree on. Executives debated methodology while customers churned. There was no standard way to answer 'Are our customers happy?' so nothing changed.

After NPS: the Net Promoter Score

Fred Reichheld reduced customer loyalty to a single question — 'How likely are you to recommend us?' — scored 0 to 10. The result: Apple's NPS exceeded 70 when competitors sat at 30. Bain & Company linked a 5-point NPS increase to revenue growth of 25 to 95%. Hundreds of companies restructured their entire customer experience operation around this one number. The score created urgency, created benchmarks, and created a clear fix-it loop.

The SEO score story: PageRank became a \$200B industry

Before SEO scoring (pre-2000)

Website owners had no idea why some pages ranked and others did not. Google's algorithm was a black box. Marketing teams spent budgets on advertising because organic search felt uncontrollable and unmeasurable.

After SEO scoring tools (Moz, SEMrush, Ahrefs)

Tools like Moz introduced Domain Authority — a 0 to 100 score predicting search ranking. Suddenly every website had a measurable problem and a clear direction. The global SEO

industry grew from near zero to over \$80 billion by 2023. Businesses hired SEO teams, agencies built practices, and vendors sold solutions — all because a score made an invisible problem visible.

The TypeOps Score opportunity

Brand typography consistency is today where customer satisfaction was in 2002 and SEO was in 1999 — a real, costly problem that nobody has a standard way to measure. Enterprise brands spend millions on typefaces and brand guidelines, yet have no automated way to know if those guidelines are being followed. The TypeOps Score changes that.

The core insight

If you can measure it, you can sell the fix. A brand team that sees 'your TypeOps Score is 38/100 — 14 unauthorised fonts detected across 6 subdomains' will immediately want to act. Monotype Connect is the action.

Problem Statement

Enterprise brands have no way to automatically measure whether their typography guidelines are being followed — across their website, marketing documents, and internal files.

When a brand cannot measure the problem, it cannot fix it — and it cannot justify the tools that would help. The TypeOps Score makes typography health measurable, automatable, and actionable.

The specific pain points

- A brand team has no automated way to detect unauthorised fonts on their website — they find out during a quarterly manual audit, months after the problem started.
- PDFs and PowerPoint decks shared publicly or stored in SharePoint contain legacy typefaces from 3+ years ago — invisible to web font audits.
- There is no industry benchmark — a brand cannot compare their typography consistency against competitors.
- Without a score, there is no urgency. Without urgency, there is no budget for solutions like Monotype Connect.

Who feels this pain

Audience	Why they care
Enterprise Brand Managers	Responsible for brand consistency across all touchpoints — currently flying blind
Legal & Compliance Teams	Font licenses can be breached unknowingly — financial and legal risk
Accessibility Teams	Untagged PDFs and low-contrast text are WCAG failures — regulatory risk
Marketing & Design Agencies	Clients ask for brand consistency — agencies have no tool to prove it
CTO / Engineering Leads	Hardcoded font values in CSS are technical debt — no current way to quantify it

The Solution: TypeOps Score Tool

A publicly accessible web tool that scans any brand's digital presence — website, public PDFs, SharePoint documents — and returns a composite TypeOps Score from 0 to 100, broken down into five weighted metrics.

The five metrics

Metric	Weight	What it measures	Why it matters
Brand Consistency	20%	Are approved fonts used across all pages and documents?	Core brand identity — the primary selling point
License Compliance	30%	Are fonts deployed within Monotype license terms?	Legal and financial risk for the brand
Web Performance	20%	Do fonts hurt Core Web Vitals (LCP, CLS)?	SEO ranking and page speed — tangible revenue impact
Accessibility	15%	Do fonts and documents meet WCAG 2.1 AA?	Regulatory risk — ADA, EN 301 549
Developer Experience	15%	Are fonts managed via tokens or hardcoded in CSS?	Technical debt and maintainability risk

Score interpretation

Score range	Status and recommended action
75 to 100 — Healthy	Typography is well-managed. Monotype Connect will help maintain and scale it.
50 to 74 — Needs Attention	Gaps identified. Addressing them improves brand consistency and reduces risk.
0 to 49 — At Risk	Significant violations. Immediate action required. Monotype Connect is the solution.

Milestones & Features

The build is structured into three phases. The hackathon targets Phase 1 completion — a working scanner with real scores on a live website.

Phase 1 — Hackathon MVP (target: demo day)

Goal

A working URL scanner that produces a real TypeOps Score for any public website.

1. User enters a website URL into the tool
2. Headless browser (Playwright) crawls up to 5 key pages
3. CSS is extracted and all font-family values are collected
4. Lighthouse API runs to capture Core Web Vitals and font loading behaviour
5. axe-core runs for accessibility checks on each page
6. All PDFs linked from crawled pages are downloaded and parsed for font names and accessibility tags
7. Scores are computed for each metric and weighted into a composite score
8. Results page shows: overall score, metric breakdown, specific violations, and a Monotype Connect CTA

Milestone acceptance: The tool must produce a non-trivial, accurate score for at least three real brand websites demonstrated on demo day.

Phase 2 — Public Launch (post-hackathon)

Goal

A polished, publicly shareable tool with lead capture and industry benchmarks.

- Scan up to 20 pages per domain
- Email gate: full breakdown unlocked after email submission
- Shareable score report as a PDF and public link
- Industry benchmark comparison ('Your score vs. Fortune 500 average')
- Microsoft SharePoint integration via OAuth — scan internal documents
- Re-scan history: track score improvement over time
- Automated lead routing to Monotype Connect sales team via HubSpot

Phase 3 — Growth & Monitoring (3 to 6 months)

Goal

Continuous monitoring product with embeddable badge and enterprise connector.

- Weekly automated re-scans with score-drop alerts via email

- Embeddable badge: 'TypeOps Score: 84 — Powered by Monotype'
- Azure AD enterprise connector for tenant-wide SharePoint scanning
- API for agencies to white-label the scanner for client reporting
- CI/CD GitHub Action: block pull requests containing unauthorised font-family values

Implementation Specification

Technical architecture

Layer	Technology & rationale
Frontend	Next.js + Tailwind CSS — fast to ship, Vercel deploy in one click
Web crawler	Playwright (headless Chromium) — full JS execution, captures fonts loaded via JavaScript
Performance audit	Google Lighthouse API — free, returns CWV metrics and font-display values
Accessibility audit	axe-core — open source, WCAG 2.1 AA checks, runs in Playwright context
PDF parsing	pdfjs-dist (Node) — extracts font dictionaries and document tags from PDF files
Office file parsing	Unzip DOCX/PPTX and parse XML — font names in word/theme/theme1.xml
Job queue	BullMQ + Redis — async scan jobs, prevents timeout on slow sites
Database	PostgreSQL — stores scan results, score history, and user/lead records
SharePoint API	Microsoft Graph API v1.0 — list files, download content, read metadata
Lead capture	HubSpot Forms API — email capture triggers CRM contact creation
Hosting	Vercel (frontend) + Railway (API + workers) — zero-config, scales for demo

Scoring algorithm — metric-level detail

Metric	Input signals	Scoring logic	Score range
Brand Consistency	Font-family values from CSS + document XML	Start at 100. Deduct 8 points per unique non-brand font found. Deduct 5 if email templates use system fonts.	0 to 100
License Compliance	Font CDN origin URLs from network requests	Start at 100. Deduct 20 per font served from an unknown CDN. Deduct 15 if estimated traffic exceeds license tier.	0 to 100
Web Performance	Lighthouse LCP, CLS, font-display value, font file sizes	Use Lighthouse Performance score as base. Apply -10 if font-display is not swap. Apply -15 if font files exceed 100KB each.	0 to 100
Accessibility	axe-core violations + PDF tag flags + contrast ratios	Start at 100. Deduct 10 per critical axe violation. Deduct 15 per untagged PDF. Deduct 10 if font size below 12px found.	0 to 100

Developer Experience	Count of hardcoded font values vs CSS variables in stylesheets	Start at 100. Deduct 3 per hardcoded font-family declaration. Deduct 20 if no CSS custom property font tokens found.	0 to 100
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Acceptance criteria — what must be true on demo day

- The tool accepts any public URL and returns a score within 60 seconds
- Brand Consistency and Web Performance scores are computed from real crawl data — not mocked
- At least one PDF found on the scanned site is parsed and its fonts are reflected in the Brand Consistency score
- The results page shows specific violations with file/page-level detail, not just a summary number
- The Monotype Connect CTA is present on the results page and links to a relevant landing page
- The tool handles a Cloudflare-protected site gracefully — shows a partial score or a clear error, not a crash

Success metrics

Metric	Target
Scan completion rate	90%+ of submitted URLs return a score (not error)
Scan time (p90)	Under 60 seconds for a 5-page crawl
Score accuracy	Brand Consistency score matches a manual font audit within 15 points
Demo day conversion	At least 3 hackathon judges express interest in using the tool
Lead quality (Phase 2)	Email captures include at least 30% enterprise domain addresses

Key risks and mitigations

Risk	Severity	Mitigation
Scan takes over 60 seconds	High	Use async job queue with live progress bar. Show partial results as each page completes. Cap crawl at 5 pages for MVP.
JavaScript-heavy SPAs miss fonts	Medium	Playwright with full JS execution and a 3-second wait-for-network-idle. Capture computed CSS, not just static HTML.
Cloudflare / bot protection blocks crawl	Medium	Add polite crawl delay (1s between pages), rotate user agents, respect robots.txt. Show graceful error if blocked.
Font file too large to download in time	Medium	Cap PDF size at 10MB, cap PDF count at 10 per domain. Show 'X PDFs not scanned — file size limit' in results.
SharePoint OAuth blocked by IT admin consent	Medium	Fall back to shared-link paste for demo day. Full OAuth is Phase 2.

Why This Idea Will Win the Hackathon

The strongest hackathon ideas share three qualities: they solve a real problem, they demonstrate value in a live demo, and they create a clear path to a commercial product. TypeOps Score has all three.

It is demonstrable in real time

On demo day, the team can scan a judge's company website live and show them their TypeOps Score. A personalised, real score with specific findings is infinitely more compelling than a slide deck. The product sells itself in the demo.

It follows a proven playbook

NPS turned 'customer satisfaction' from a vague feeling into a number — and built a billion-dollar consulting industry. SEO scores turned 'search visibility' from a black box into an actionable metric — and built an \$80 billion market. TypeOps Score does the same for brand typography. The playbook is proven. The market is ready.

It is a natural top-of-funnel for Monotype Connect

Every brand that scans and gets a low score is a warm lead. They have seen their problem quantified, with file-level evidence. The Monotype Connect pitch writes itself: 'Your score is 38. Here is how Connect fixes each of these violations.' This is not a hackathon toy — it is a demand generation tool.

The one-line pitch

TypeOps Score is the NPS for brand typography — one number that tells enterprise teams exactly how consistent, compliant, and accessible their fonts are across every digital surface, and shows them how to fix it with Monotype Connect.